

DERRELL BLACK

AI CONSULTANT · AUTOMATION ARCHITECT
· EDUCATOR

(312) 965-2782

Derrellblack@BlacktechSolutionscorp.com

Chicago, IL · Available Remote & On-Site

PROFILE

AI consultant and automation architect with hands-on experience building production AI systems on edge hardware. Founded an AI education academy with 8 courses spanning AI, blockchain, and crypto. Specialize in deploying LLM agents, automated workflows, and self-hosted infrastructure for small businesses.

CONTACT

Phone: (312) 965-2782

Email:

Derrellblack@BlacktechSolutionscorp.com

Location: Chicago, IL 60628

Availability: Remote & On-Site

TECHNICAL SKILLS

- ▶ AI Agent Development (Hermes, OpenAI, Ollama)
- ▶ LLM Integration & Prompt Engineering
- ▶ Python · Bash · JavaScript · HTML/CSS
- ▶ SQLite · PostgreSQL · API Development
- ▶ Raspberry Pi / Edge Computing
- ▶ Linux System Administration
- ▶ Docker & Container Orchestration
- ▶ Cloudflare Tunnels & DNS Management
- ▶ Web3 / Blockchain (Solidity, Smart Contracts)
- ▶ Video Generation & TTS Pipelines
- ▶ Web Scraping & Data Pipelines
- ▶ Cron Automation & Scheduled Workflows

CERTIFICATIONS

- ✓ ICC EESP — Energy Efficiency Service Provider
- ✓ ICC EV — Electric Vehicle Charging
- ✓ ICC DG — Distributed Generation
- ✓ MBE Certified — Minority Business Enterprise
- ✓ *Google AI Essentials (in progress)*
- ✓ *Microsoft Azure AI Fundamentals AI-900 (planned)*
- ✓ *DeepLearning.AI Specialization (planned)*

EDUCATION

Coyne American Institute
1998
Electrical Maintenance

NECA/IBEW 134 Technical Institute
2010
Communication

AI & CONSULTING EXPERIENCE

Independent AI Consultant CURRENT 2025 — Present
AI Infrastructure & Automation Consulting

Design and deploy AI-powered automation systems for small businesses. Build self-hosted infrastructure on edge hardware (Raspberry Pi) that replaces costly SaaS subscriptions.

- Deployed 25+ always-on services on a single Raspberry Pi with automated monitoring, restart, and alerting
- Built AI agent system (Hermes) with 50+ automated workflows handling lead generation, invoicing, social media, and business operations
- Created voiceover video generation pipeline using TTS + slideshow automation
- Positioned consulting practice: "I built this for myself — here's what I can build for you"

Founder & Curriculum Director CURRENT 2024 — Present
AI Education Academy

Founded an online AI education platform with 8 courses covering AI fundamentals, blockchain, cryptocurrency trading, and Web3 development. Built the entire platform from scratch — server, database, video pipeline, and course content.

- Authored 8 courses with 30+ modules and 75+ video lessons
- Built self-serve registration, progress tracking, and gamification (badges, token rewards)
- Integrated blockchain token reward system for course completion
- Distributed AI education flyers to public libraries for community outreach

Energy Advisor & Technology Lead 2021 — Present
Energy Solutions · Think Energy Partner

Combined 28+ years of electrical expertise with AI automation to modernize energy consulting. Built automated lead pipelines, contractor databases, and utility rebate calculators.

- Built CRM with 3,489+ contractors, automated lead scoring, and dispatch board
- Created automated lead generation across 6 sources (Nextdoor, Craigslist, ComEd, Apollo, DDG, BBB)
- Deployed AI-powered bid monitoring system scanning government and commercial bid portals
- Automated ComEd utility rebate calculations and energy savings estimates

KEY AI PROJECTS

AI Agent Automation Platform

Production AI agent system running 50+ scheduled workflows: lead generation, social media posting, email monitoring, invoice generation, and business intelligence — all on edge hardware.

[Python](#) · [Ollama](#) · [SQLite](#) · [Cron](#) · [Telegram API](#) · [Cloudflare](#)

Command Center (25+ Services)

Unified web dashboard managing 25+ always-on services: CRM, dispatch board, academy, search engine, crypto trading platform, video pipeline, and automated monitoring.

[Python](#) · [nginx](#) · [Cloudflare Tunnels](#) · [Docker](#) · [systemd](#)

AI Video Generation Pipeline

Automated slideshow + AI voiceover video creation pipeline. Converts scripts into narrated video presentations with background music, deployed to streaming platforms.

[Python](#) · [Edge-TTS](#) · [ffmpeg](#) · [HTML/CSS](#) · [Chromium headless](#)

Web3 Token System

Built a custom ERC-20 token with smart contracts on Sepolia/Base testnets. Integrated token rewards into education academy and trading platform.

[Solidity](#) · [Web3.py](#) · [Smart Contracts](#) · [MetaMask](#) · [Python](#)

Community Lead Generation Engine

Automated lead scraping across multiple sources (5,000+ church leads, 3,489 contractors), with qualification scoring and weekly digest delivery.

[Python](#) · [Playwright/Camoufox](#) · [SQLite](#) · [Cloudflare bypass](#)